

5.1.2 Internet of Things (IoT) & Components

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں...

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 03 of 12

★ Golden

5.1.2 Internet of Things (IoT) & Components

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LEARN IT

The **Internet of Things (IoT)** is a system in which **physical devices are connected to the Internet** and can **collect and share data** to perform tasks **automatically**.

The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so they can "talk" to each other and to us.

Source example: A **smartwatch** collects your heart-rate data and sends it to your mobile phone, where it is displayed.

Components of IoT systems

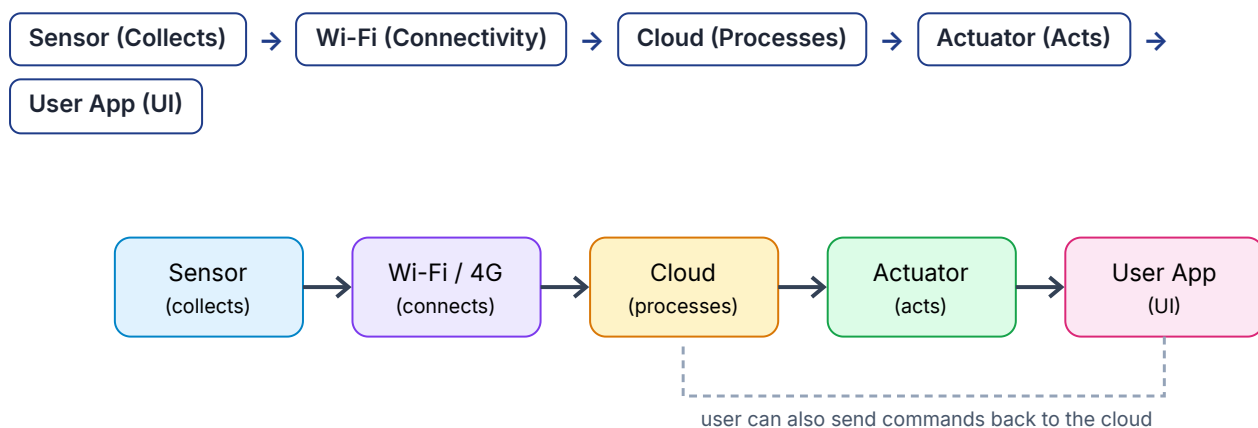
IoT relies on **six foundational components** to work properly.

#	Component	What it does	Examples
1	Sensors	Devices that collect information from the physical world. They are the "eyes and ears" of the system.	Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor
2	Actuators	Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.	Electric motor, switch, automatic door opener, water valve, fan controller
3	Connectivity	The connection that allows devices to send and receive data . Without it, sensors can't talk to the system.	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	Data Processing	Changing raw data into useful information. Steps: collecting, organising, analysing and deciding.	Checking if the room temperature is too hot to trigger the AC
5	Cloud Computing	Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.	Google Drive, iCloud, online LMS
6	User Interface (UI)	The component that enables interaction between the user and the device . Helps users see information and give commands.	Mobile apps, touch screens, voice commands

Understanding each component

- **Sensors** turn something in the real world (heat, movement, a fingerprint, smoke, wetness of soil) into data. A smoke sensor in a Karachi shopping mall "smells" smoke and reports it.
- **Actuators** do the opposite: they take a digital command and make something *move* or *change*. The water valve that opens, the motor that turns, the door that slides open at a bank entrance.
- **Connectivity** is the "road" for data. Short distances use Bluetooth (watch to phone); a house uses Wi-Fi; low-power sensor networks use **ZigBee**; devices far away in fields use mobile networks (4G/5G from Jazz, Zong, Telenor/Ufone).
- **Data processing** is where raw numbers ("36") become meaning ("too hot — turn on AC"). Its four steps are *collect* → *organise* → *analyse* → *decide*.
- **Cloud computing** stores and processes the huge amount of data that thousands of sensors produce every second — far more than a single small device can hold.
- **User Interface** is how *you* see the data and give orders — an app on your phone, a touch screen on a smart fridge, or saying "Hey Google, turn off the lights".

How the components connect



The six IoT components working as one system (data processing happens inside the cloud).

WORKED EXAMPLE

Smart fire alarm in a Karachi factory — identify all six components:

Step 1 — Sensor: A smoke sensor detects smoke near the machines.

Step 2 — Connectivity: It sends the reading over the factory Wi-Fi.

Step 3 — Cloud computing: The reading is stored on an online server along with the history of all sensors.

Step 4 — Data processing: The software checks: "Is the smoke level above the safe limit?" Yes → decide "Fire risk!"

Step 5 — Actuator: Water valves of the sprinklers open and the alarm bell switch turns on.

Step 6 — User interface: The manager's mobile app shows a red alert: "Smoke detected – Hall 3".

REMEMBER IT

Sensor senses, Actuator acts. Both words give you the answer: **sensor** = **see**/sense (input), **actuator** = **act** (output). Remember the six components with "**Some Ants Carry Data Carefully Up**" — Sensors, Actuators, Connectivity, Data processing, Cloud, UI.

COMMON MISTAKE

Mistake: Calling a fan controller or water valve a "sensor". **Fix:** Ask "does it *measure* something or *do* something?" A valve opens and closes — it *does*, so it is an **actuator**. A soil moisture sensor only *measures*, so it is a **sensor**.

اردو میں سمجھیں

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے ڈیٹا جمع اور شیئر کرتے ہیں۔ اس کے چھ بنیادی اجزاء ہیں: سینسرز (معلومات جمع کرتے ہیں)، ایکچویٹرز (جسمانی کام انجام دیتے ہیں جیسے موٹر چلانا یا دروازہ کھولنا)، کنیکٹیویٹی (وائی فائی، بلوٹوتھ)، ڈیٹا پروسیسنگ، کلاؤڈ کمپیوٹنگ، اور یوزر انٹرفیس (موبائل ایپ)۔

EXAM TIP

"Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — *physical devices, connected to the internet, collect and share data automatically*. Then list all **six** components, each with a one-line meaning and one example. A labelled flow diagram (Sensor → Wi-Fi → Cloud → Actuator → App) earns extra credit.

Short questions often ask: "Differentiate between a sensor and an actuator" — sensor collects data (input, eyes and ears), actuator performs physical action (output, converts digital signals into movement).

CHECK YOURSELF

1. Classify as sensor or actuator: (a) fingerprint scanner, (b) automatic door opener, (c) motion detector, (d) electric motor.
2. Name four types of connectivity used in IoT.
3. Why is cloud computing needed in IoT?

Answers: 1. (a) sensor, (b) actuator, (c) sensor, (d) actuator. 2. Wi-Fi, Bluetooth, ZigBee, 4G/5G. 3. Because IoT devices produce massive amounts of data; the cloud stores and processes it over the internet instead of on small local hard drives.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. IoT is:

- a. Physical devices on the internet sharing data
- b. Only desktop Word
- c. Only K-maps
- d. Only sprints

MCQ 2. A typical IoT chain is:

- a. Sensor → connectivity → data → action
- b. Query → form → report only
- c. SYN → ACK only
- d. Bubble → selection

MCQ 3. Connectivity might be:

- a. Wi-Fi, 4G, Bluetooth
- b. A primary key
- c. A pie chart
- d. A NAND-only bus

MCQ 4. An actuator:

- a. Does something in the real world (motor, relay)
- b. Only stores SQL
- c. Only sorts
- d. Only prints Urdu

MCQ 5. Without connectivity, sensors:

- a. Cannot send readings
- b. Become RDBMS
- c. Become AI models
- d. Become OSI layer 7

MCQ 6. A smart irrigation valve is IoT because:

- a. It senses, sends, and acts remotely
- b. It is made of paper
- c. It runs bubble sort
- d. It is a form wizard

Extra short questions

1. Define IoT.
2. List four components.
3. Give a farm example.
4. One security worry.

Extra long question

1. Draw the IoT component diagram for a smart classroom (lights, attendance, AC). Label each component and one risk. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Internet of Things: physical objects with sensors/software that connect and exchange data.
2. Devices/sensors, connectivity, data processing, user interface / actuators.
3. Soil moisture sensor → mobile network → app → pump on/off.
4. Unpatched cameras on the internet; weak default passwords.

Long

1. Sensors (PIR, RFID), Wi-Fi, college server/analytics, dashboard, relays for lights. Risk: student location privacy.
Mitigation: local processing, access control.